Name : ________________ Class : ________ ID : ______________

1 What is carbon steel?

ANS: Steel is iron alloy with carbon alloying element. In other words, carbon steel alloys come in many compositions but there will always have Fe and C.

2 What is cold working?

ANS: Cold working refers to the process of strengthening metal by changing its shape without the use of heat. Subjecting the metal to this mechanical stress causes a permanent change to the metal's crystalline structure, causing an increase in strength.

3 We have the following alloying elements for steel alloy:-

- Nickel
- Chromium
- Molybdenum

Please advise what benefits come from having these respective alloying elements adding into the carbon steel alloy.

ANS:

- Nickel:- Typically increases the hardness, tensile strength, and elastic limit of steel without appreciably decreasing the ductility.
- Chromium:- Chromium steel alloy is typically high in hardness, strength, and corrosion resistant properties and is particularly adaptable for heat-treated forgings, which require greater toughness and strength.
- Molybdenum:- It raises the ultimate strength of steel without affecting ductility or workability. Molybdenum steels are tough and wear resistant, and they harden throughout when heat-treated. They are especially adaptable for welding and, for this reason, are used principally for welded structural parts and assemblies.

4 How is heat treatment is done to change the internal structure of the steel alloys.

ANS:

- heating to a temperature above its upper critical point
- holding it at that temperature for a time sufficient to permit certain internal changes to occur
- and then cooling to atmospheric temperature under predetermined, controlled conditions.

5 Define the following heat treatment:-

- Hardening
- Tempering
- Annealing
- Normalising

ANS:
Heat treatment depends on the carbon content of the steel alloy itself and the method of heating, soaking and cooling.

- To harden a steel alloy, it involves heating the steel to a temperature just above the upper critical point, soaking or holding for the required length of time, and then cooling it rapidly by plunging the hot steel into oil, water, or brine. Although most steels must be cooled rapidly for hardening, a few may be cooled in still air.
- Tempering reduces the brittleness imparted by hardening and also softens the steel. Tempering always following after hardening is done. The hardened steel alloy is reheated to a temperature between the lower critical point and 212 °F / 100 °C and hold it there for a defined period to achieve the hardness and strength determined.
- Annealing involves heating the steel alloy to just above the upper critical point, soaking at that temperature, and cooling very slowly in the furnace accomplishes annealing of steel. Annealing of steel produces a coarse-grained, soft, ductile metal without internal stresses or strains which also increase in toughness. When compared to a harden steel alloy, please note that it is complete removal of all stresses, coarse grains (large grains), toughness is increased, ductility is increased, tensile strength is not affected or only slightly improved.
- The normalizing of steel removes the internal stresses set up by heat treating, welding, casting, forming, or machining. It involves heating the steel alloy above the upper critical point and cooling in still air. This is similar to annealing except the quenching is more rapid resulting in a harder and stronger material. Please note that normalizing typically applies to iron base metals only.

6 Why are some components casehardened instead of fully harden?

ANS: Casehardening produces a hard, wear-resistant surface or case over a strong, tough core. Casehardening is ideal for parts that require a wear-resistant surface and, at the same time, must be tough enough internally to withstand the applied loads.

7 Why is aluminium is one of the most widely used metals in modern aircraft construction?

ANS: Aluminium is vital to the aviation industry because of its high strength-to-weight ratio and its comparative ease of fabrication.

8 What does the terms “Alclad” and “Pureclad” mean?

ANS: The terms “Alclad and Pureclad” are used to designate sheets that consist of an aluminum-alloy core coated with a layer of pure aluminum to a depth of approximately 5½ percent on each side. The pure aluminum coating affords a dual protection for the core, preventing contact with any corrosive agents, and protecting the core electrolytically by preventing any attack caused by scratching or from other abrasions.

9 Describe hardening of aluminium alloy through heat treatment.

ANS:
This is very similar to tempering of steel except that the goal is to increase both the strength and hardness of the metal alloy. There are two types of heat treatments applicable to aluminum alloys: **solution heat treatment** and **precipitation heat treatment**.
The hardening of an aluminum alloy by heat treatment consists of four distinct steps:

- **1. Heating to a predetermined temperature.**
- **2. Soaking at temperature for a specified length of time.**
- **3. Rapidly quenching to a relatively low temperature.**
- **4. Aging or precipitation hardening either spontaneously at room temperature, or because of a low temperature thermal treatment.**

The **first three steps** above are known as solution heat treatment, although it has become common practice to use the shorter term, **“heat treatment.”**

Step 4 is for aging or precipitation heat treatment. The alloys that require precipitation heat treatment (artificial aging) to develop their full strength also age to a limited extent at room temperature; the rate and amount of strengthening depends upon the alloy.

10 What are the differences between heat treatment for steel alloy (carbon) and non-steel alloy*

ANS: There is not much difference if we look at the processes themselves. The processes such as the critical temperature, soaking duration, and cooling rates….. are determined by the alloy compositions themselves.
Also the compositions also define the internal structures and what can be precipitated out to improve hardening or strengthening of the alloys.

11 Metallic aircraft uses mostly 2000 and 7000 series of aluminium for the fuselage and wing respectively. Explain why so.

ANS: Aluminium alloys are known for their high-strength to weight ratio making it a suitable material for primary structure with cost considered.
2000 and 7000 series aluminium alloys are known for their high strength properties and the two series are able to undergo heat treatment to increase its strength and hardness accordingly.

12 6000 series aluminium is used for the bilge of the fuselage. Explain why so.

ANS: Aluminium alloys are known for their high-strength to weight ratio making it a suitable material for primary structure with cost considered. However, the bilge area is corrosion prone area, thus to prevent corrosion, the most corrosion resistance aluminium was chosen. Furthermore, this series is able to undergo heat treatment to increase its strength and hardness accordingly.